

ECO 321—ECONOMETRICS

Spring 2018

Department of Economics
Stony Brook University

Instructor: Alejandro Melo Ponce

Lecture time: Mondays & Wednesdays, 4:00 PM–5:20 PM

Classroom: Humanities 1003

Office hours: Wednesdays 9:00 AM–12:00 PM SBS N604

E-mail: alejandro.meloponce@stonybrook.edu

Graduate TA: Dong Wook Kang

Recitations: Mondays, 1:00 PM–1:53 PM (R04) & 2:30 PM–3:23 PM (R03)

Classroom: Earth and Space Sciences SINC site (Room 081)

Office hours: Mondays 10:00 AM–1:00 PM SBS S630

E-mail: dongwook.kang@stonybrook.edu

Undergraduate TAs: Xuliejun Ren & Yundai Hu

Office hours: Thursdays 5:00 PM–7:00 PM (Xuliejun) & Tuesdays 4:00 PM–6:00 PM (Yundai), Economics Department Lobby, SBS 6th floor

E-mail: xuliejun.ren@stonybrook.edu & yundai.hu@stonybrook.edu

Course Description

This class will introduce you to statistical techniques used by applied econometricians to analyze real world phenomena. In this class, theory will be motivated by discussing applications. First, we will review some core ideas in probability theory that are necessary to understand regression analysis and econometrics. Second, we will cover regression analysis and OLS estimation, hypothesis testing, and extensions to the linear regression model. Finally, we will discuss topics in model selection and evaluation. If time permits, we will cover binary dependent variable models, endogeneity issues and instrumental variables.

Course Requirements

C or higher in ECO 320 or AMS 310; C or higher in ECO 108

Course Material

- **Textbook (Required):** *MyLab Economics with Pearson eText for Introduction to Econometrics* by Stock, James H. and Watson Mark W., Pearson, 2011, 3rd Edition, ISBN-13: 978-0133487671. This is the required material for the course. This will provide access to the MyLab online platform in which you will deliver the weekly quizzes. Furthermore, the platform provides access to an eText version of the book on which this course is based and from which you will be required to do the readings for the class. You can buy it directly from Pearson by logging first into the BlackBoard course web page and then clicking on the MyLab link. That will lead you to the MyLab Pearson website where you will be able to choose the option that is most convenient to you.

- **Textbook (Recommended):** *Introductory Econometrics: A Modern Approach* by Jeffrey Wooldridge, Cengage Learning, 2016, 6th Edition, ISBN-13: 978-1-305-27010-7. A more formal introduction to Econometrics. The presentation, however, is somewhat different from Stock and Watson. Of particular interest is the mathematics, probability and statistics review in the appendix.
- **Textbook (Recommended):** *Mastering 'Metrics, The path from cause to effect* by Joshua D. Angrist and Jorn-Steffen Pischke, Princeton University Press, 2014, 1st Edition, ISBN-13: 978-0-691-15284-4. A very cheerful and enjoyable account of applied econometrics. It has a superb discussion of causal inference and an extremely accessible and enlightening discussion of instrumental variables.
- **Supplementary materials and lecture notes** will be posted on Blackboard: <https://blackboard.stonybrook.edu/>. You are responsible for checking for announcements on BB regularly.

Attendance

Attending lectures is *highly* recommended. By attending lectures you will be able to benefit from in-class presentations and discussions. You are also responsible for learning everything discussed in class whether these topics are in the textbook or not.

Rules and Regulations

During lectures, please keep your cell phones switched off and refrain from text messaging. Inattentive students whose actions are judged to be disruptive to class discussion will be asked to leave.

Grading Policy

Your final grade for the course consists of the following items:

- BlackBoard quizzes: 20%
- Problem sets: 40%
- Midterm 1: 20%
- Midterm 2: 20%

See the sections below for more information.

After computing your weighted average, the letter grade distribution will be as follows.

Letter Grade	Score Achieved
A	[93, 100]
A—	[90, 93)
B+	[87, 90)
B	[83, 87)
B—	[77, 83)
C+	[70, 77)
C	[63, 70)
C—	[57, 63)
D+	[50, 57)
D	[40, 50)
F	[0, 40)

BlackBoard/MyLab Quizzes

You will have weekly quizzes that you will access through BlackBoard and will be submitted using the MyLab platform. These quizzes are intended to cover your basic knowledge of the topics covered during that week's lecture. They are an important stepping stone in assessing if you are ready to tackle the problem sets. Each week the quiz will be posted on BlackBoard/MyLab after Wednesday's lecture starting from the second week and is due on the following Saturday at 11:00 PM. For your final grade, I will drop your *three worst quiz grades*, which will not count towards your final score. There will be **no makeup quizzes** so you need to adhere strictly to the deadline.

Problem Sets

There will be six problem sets. In each problem set you will have some theoretical and practical exercises as well as some applied exercises. The applied exercises involve real world data sets in which you will have to use the theory learned during the lectures to perform some econometric inference. You will need to solve this part using the statistical software R. You will learn how to use this software during recitations.

Problem Sets will be due in class and no late submissions will be accepted **unless a valid reason is provided**. For your final grade, I will drop your *worst problem set grade*, which will not count towards your final score. The deadline for submitting the problem sets is strict, so please check the due dates in the schedule below.

Midterms

There will be two midterms in the course. None of the exams is cumulative. The first midterm is in the week before Spring Break and will cover the first half of the course. The second midterm is on the final day of classes and will cover the second half of the course. You can find the dates of each midterm in the schedule below. Midterms 1 and 2 will be during normal lecture time. **There will be no make-up exams unless a valid reason is provided.**

Final Exam (Optional)

There will be an optional final exam, to be held on Tuesday May 9, 2018 from 8:30PM–11:00PM per the Registrar's schedule. The final exam can be used to replace the grade of one of the midterms. If you decide to take the final exam, the grade you get will be compared to your **lowest midterm grade**. If the score of the final is higher, then I will replace your lowest midterm grade with that score.

There is no need to take the final if you don't want to take advantage of this opportunity. The final, in contrast with the midterms, will be **cumulative**, i.e. will cover all the course contents.

Recitations

Every week this class will have a recitation session conducted by your TA. As detailed in the course description, an important part of Econometrics is understanding how to apply the theory to analyze real world phenomena. In particular, is very important that you become familiar with how to handle data by using statistical software.

During these recitation sessions your TA will teach you how to use the statistical software package R. R is a *free* software environment for statistical computing and graphics. It is very important that you attend the recitation sessions, not only because you must become acquainted during this course with handling data and with statistical software but also because each one of your problem sets will include an applied component which you must solve using R.

Important Dates

Please, check assignment submission and exam dates. Since I drop your three lowest grade quizzes and your lowest grade problem set, the submission deadlines for these are **strict**. **There will be no makeup quizzes, problem sets, or exams unless a valid reason is provided.** See below for more information. If you have a schedule conflict that prevents you from taking the midterms on the scheduled date, please contact me as soon as possible so we can make arrangements.

Tentative Course Outline

The weekly coverage might change as it depends on the progress of the class. Changes, if any, will be announced in class.

Week	Class	Topic	Reading	Recitation
Week 1	1/22	Introduction to the Course	SW 1	No Recitation
	1/24	Review of Probability.	SW 2	
Week 2	1/29	Review of Probability.	SW 2	Recitation 1
	1/31	Review of Probability.	SW 2	
Week 3	2/5	Review of Statistics. Assignment 1 due.	SW 3	Recitation 2
	2/7	Review of Statistics.	SW 3	
Week 4	2/12	Review of Statistics.	SW 3	Recitation 3
	2/14	Single Linear Regression Model. Assignment 2 due.	SW 4	
Week 5	2/19	Single Linear Regression Model: Measures of Fit	SW 4	Recitation 4
	2/21	Single Linear Regression Model:Hypothesis Tests and Confidence Intervals	SW 5	
Week 6	2/26	Single Linear Regression Model:Hypothesis Tests and Confidence Intervals	SW 5	Recitation 5
	2/28	Homoskedasticity and Heteroskedasticity. Gauss Markov Theorem. Assignment 3 due.	SW 5	
Week 7	3/5	Review Session		Recitation 6
	3/7	Midterm 1.		
		Spring Break		
Week 8	3/19	Linear Regression with Multiple Regressors. Omitted Variables.	SW 6	Recitation 7
	3/21	Linear Regression with Multiple Regressors. Foundations	SW 6	
Week 9	3/26	Linear Regression with Multiple Regressors. Measures of fit and fundamental assumptions. Assignment 4 due.	SW 6	Recitation 8
	3/28	Hypothesis Tests in the Multiple Regression Model.	SW 7	
Week 10	4/2	Hypothesis Tests in the Multiple Regression Model. Testing joint hypothesis	SW 7	Recitation 9
	4/4	Hypothesis Tests in the Multiple Regression Model. Restrictions involving multiple coefficients.	SW 7	
Week 11	4/9	Non-linear regression functions. Polynomials, logarithms and interaction terms. Assignment 5 due.	SW 8	Recitation 10
	4/11	Non-linear regression functions: Interaction terms. Model Validity.	SW 8, 9	
Week 12	4/16	Regression with Binary Dependent Variables. The probit and logit model.	SW 11	Recitation 11
	4/18	Regression with Binary Dependent Variables: Inference. Assignment 6 due.	SW 11	
Week 13	4/23	Instrumental Variables: Endogeneity Problem. The IV estimator with a single regressor and a single instrument.	SW 12	Recitation 12
	4/25	The IV estimator in the general model.	SW 12	
Week 14	4/30	Review Session. Assignment 7 due.		Recitation 13
	5/2	Midterm 2.		
	5/8	Final Exam (Optional)		

Absentee Policy

Students are expected to report for their examinations and major graded coursework as scheduled. If a student is unable to report for any examination or to complete major graded coursework on time, the student must contact me immediately. If the student cannot reach me, then s/he should contact the Director of Undergraduate Studies.

As per Stony Brook University policy, valid reasons for absences are:

- Student Participation in University Sponsored Events.
- Equivalent Opportunity and Religious Absences.
- Significant illness, tragedy or other personal emergencies.

It is the student's responsibility to provide sufficient documentation to support any such request. Accommodations for other reasons will be at my discretion.

Special Needs

If you have a physical, psychological, medical or learning disability that may impact your course work, please contact Disability Support Services, 128 ECC Building (631) 632-6748. They will determine with you what accommodations are necessary and appropriate. All information and documentation is confidential.

Students who require assistance during emergency evacuation are encouraged to discuss their needs with their professors and Disability Support Services. For procedures and information go to the following web site: <http://www.ehs.sunysb.edu> and search Fire Safety and Evacuation and Disabilities.

Academic Integrity

Each student must pursue his or her academic goals honestly and be personally accountable for all submitted work. Representing another person's work as your own is always wrong. Faculty are required to report any suspected instances of academic dishonesty to the Academic Judiciary. For more comprehensive information on academic integrity, including categories of academic dishonesty, please refer to the academic judiciary website at http://www.stonybrook.edu/commcms/academic_integrity/index.html

Critical Incident Management

Stony Brook University expects students to respect the rights, privileges, and property of other people. Faculty are required to report to the Office of Judicial Affairs any disruptive behavior that interrupts their ability to teach, compromises the safety of the learning environment, or inhibits students' ability to learn. Faculty in the HSC Schools and the School of Medicine are required to follow their school-specific procedures.

Disclaimer

Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on BlackBoard.